

# The Ozempic Economy: Sugar Addiction, Labor Market and Optimal Subsidies

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**Christian Alemán-Pericón**

NYU Abu Dhabi

**Fane Groes**

Copenhagen Business  
School

**Veda Narasimhan**

NYU Abu Dhabi

**Raul Santaeulàlia-Llopis**

NYU Abu Dhabi,  
BSE, and CEPR

**Tetyana Surovtseva**

University of Barcelona,

VERY, VERY, VERY PRELIMINARY

# Motivation

- Obesity is a public health crisis (overweight population: 70% US, 54% Denmark)
  - ▷ Obesity negatively correlated with employment. Income Employment
- Enormous growth in semaglutides take-up (e.g. Ozempic, Mounjaro, Wegovy):
  - ▷ 500% increase: From 5 users (per 1000) in 2018— to 30 in 2023.— (Denmark)
  - ▷ Semaglutides effective at reducing weight.
- Increasing cost of subsidizing weight-loss medication.
  - ▷ 2023: Danish government spent USD200 mill. on Ozempic (8% of budget)
  - ▷ Ongoing renegotiations of drug prices (from USD188 to USD125).

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  - ▷ 2 costly technologies to lose weight:
    1. Physical exercise
    2. Semiglutides (Ozempic, Wegoby)

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  - ▷ 2 costly technologies to lose weight:
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    2. Semiglutides (Ozempic, Wegoby)
  - ▷ Price of Semiglutides is not competitive.
- OLG Model with sugar addiction, "Ozempic" and physical exercise:
  - ▷ Optimal weight-loss drug subsidy
  - ▷ Vs. a tax on processed foods (high sugar foods)

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Study the general equilibrium effects of "Ozempic" on the labor market (hours, employment, wages)



Stylized Facts: Focus on working age population

- ▷ **Today:** The obese and co-morbidities (diabetic) (Ozempic).
- ▷ **Next:** "Healthier" populations (non-diabetic) (Wegovy).

## Context

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## Semaglutides (Ozempic, Wegovi, Mounjaro): Basic Facts

- Ozempic → Developed to treat type 2 diabetes:
  - On prescription
  - Currently subsidised for diabetic patients in Denmark
- Ozempic → highly effective for weight loss
- Wegovy → introduced for weight-loss in 2022
- Semaglutides require continuous use to maintain weight loss benefits.
- Increasing interest in their broader use
  - Raises questions about reimbursement and long-term access.

# Panel Data Analysis: Semaglutides and Labor Market Outcomes

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## Our Sample: Danish Administrative Data

Quarterly panel: between q1-2014 and q2-2024

Ages: 18-65

We observe, health status (diagnosis and prescriptions/ treatment), employment status, hours worked, wages, background characteristics (age, education, marital status, gender, immigration status).

- We restrict our sample to individuals who satisfy at least one of the following conditions:
  - Have been diagnosed diabetes between 2014-2024
  - Have been diagnosed as overweight, obese, morbidly obese
  - Have been prescribed insulin
  - Have been prescribed any semaglutide

# Our Sample: Danish Administrative Data

N=850,420 individuals

- 59% are women
- Average 12 years of education
- Average age 54, Median Age 55
- 7% have been diagnosed as overweight, with 15% classified as obese
- 25% have been diagnosed with diabetes in sample period.
  - 76% have been taking insulin
  - 36% have ever taken semaglutides

## 1. Semaglutide users: Permanent vs. Temporary

- **Treatment** → Permanent users: those prescribed semaglutides, and continue treatment for  $> 3$  quarters after the first prescription.
- **Control** → Temporary users: those prescribed semaglutides, but drop treatment within 3 quarters. Do not get prescribed semaglutides again.

Alternative Empirical Strategy

# Semaglutides: Permanent Use vs. Temporary Use - Empirical Strategy

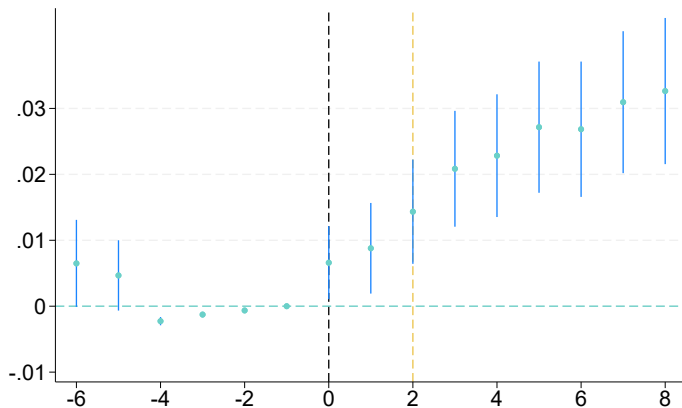
We use a difference-in-difference event study set-up:

$$y_{it} = \sum_{\tau=-6, \tau \neq -1}^8 \gamma_{\tau} D_{i\tau(it)} + \mu_{\tau(it)} + \nu_t + X_{it} + \epsilon_{it}$$

- $Y_{it}$ : employment, log earnings for individual  $i$  in quarter  $t$
- $D_{i\tau(it)}$ : equals to 1 if **treatment** (on semaglutide for 3+ quarters) and 0 if **control** (on semaglutide for fewer than 3 quarters)
- $\mu_{\tau(it)}$  is the time relative to first treatment fixed effect
- $\nu_t$ : calendar year and quarter fixed effects.
- $X_{it}$ : individual controls

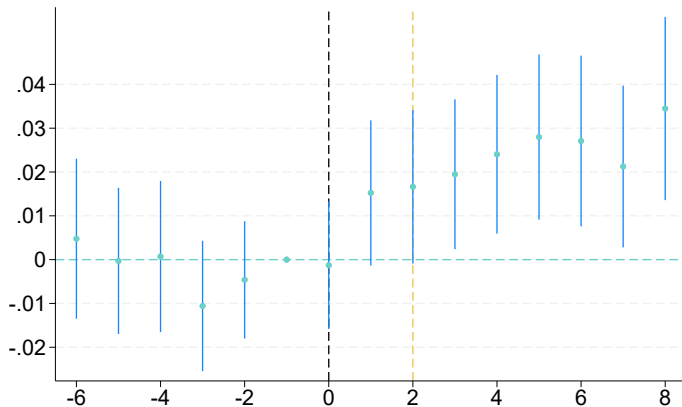


## Event Study: Increase in Employment, 2.5%



**Figure 1:** Effect on Employment

## Event Study: Increase in Earnings, 2.9%



**Figure 2:** Effect on Log Earnings

## Event Study: Increase in Hours, 5 hours

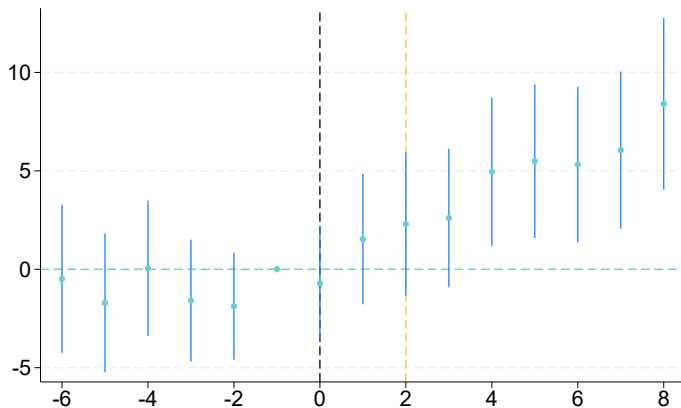


Figure 3: Effect on Hours

## An Equilibrium Model

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OLG with (new) elements:

- Consumption of processed/addictive food:  $x$
- Accumulation of sugar in blood:  $d$  (through  $x$ )
- Disutility of work that increases with BMI:  $b$
- Costly physical exercise:  $e$
- Human capital accumulation:  $s$
- Subsidized weightloss drug:  $o$

$$V(a, s, d, b, m, e_{-1}) = \max_{\{c, h, x, e, o\}} u(c, x - \underline{x}(d)) - \kappa(b)v(h) - \tilde{n}(e, e_{-1}) \\ + \beta\gamma(a', b', d', m')V(a', s', d', b', m', e)(1)$$

- $a \in [1, a_T]$  age
- $x \geq 0$  is consumption of the addictive good (processed food);
- $o$  is the amount of Ozempic purchased;
- $d$  is the stock (level) of sugar in the blood created by past processed food;
- $\underline{x}(d)$  is the habit formation of (cravings for) processed food which increases with stock of sugar in the blood  $\underline{x}_d(d) > 0$ ;
- $s$  is the stock of human capital
- $\kappa(b)$  is the disutility of work as a function of body-mass index ( $BMI$ );
- $\gamma(j, b, d, m)$  Survival rate,

# Individuals problem

$$V(a, s, d, b, m, e_{-1}) = \max_{\{c, h, x, e, o\}} u(c, x - \underline{x}(d)) - \kappa(b)v(h) - \tilde{n}(e, e_{-1}) \\ + \beta\gamma(a', b', d', m')V(a', s', d', b', m', e)(2)$$

s.t

**Budget Constraint**

$$c + p_x(\Phi)x + p_o(b)o = \underbrace{\tau(w(\Phi)h)}_{\text{Disposable Income}} \quad (3)$$

**Time Constraint**

$$1 = \ell + h + e \quad (4)$$

## Subject to: Law of motion of sugar in blood

$$d' = q^{(d)}(x) + (1 - \delta^{(d)}(o, e, m))d \quad (5)$$

- $\delta^{(d)}(o, e, m) \in (0, 1)$  depletion of sugar
- $\delta_o^{(d)}(o, e, m) > 0$  Ozempic lowers sugar in the blood, makes individuals less hungry tomorrow via a lower  $\underline{x}(d)$
- $\delta_e^{(d)}(o, e, m) > 0$ , physical activity can lower blood glucose.
- $\delta_m^{(d)}(o, e, m) > 0$ , higher muscle mass reduces blood glucose.



## Law of motion of BMI

$$b' = q^{(b)}(x) + (1 - \delta^{(b)}(e))b \quad (6)$$

- $\delta_e^{(b)}(e) > 0$  and  $\delta_{ee}^{(b)}(e) < 0$

## Law of motion of muscle mass

$$m' = q^{(m)}(a, e, o) + (1 - \delta^{(m)}(a))m \quad (7)$$

- $\delta_a^{(m)}(a) > 0$

## Human capital accumulation

$$s' = g(h) + (1 - \delta^{(s)})s, \quad (8)$$

- $g_h(\cdot) > 0$

## Population Dynamics

$$\Omega' = \gamma\Omega + (1 - \gamma)\Omega_{a=1} \quad (9)$$

$$(10)$$

## Firms Problem: Two sectors: $i = \{c, x\}$

**Firm's maximize profit:**

$$\max_h p_i Y_i - w_i \mathcal{H}_i. \quad (11)$$

**s.t**

$$p_i Y_i = p_i A_i \mathcal{H}_i$$

with  $p_c = 1$  and  $\mathcal{H}_i = \int s h_i d\omega$

**FOC( $h_i$ )**

$$w = p_i A_i s \quad (13)$$

## The government

- Collects income taxes,
- Pays for weight loss subsidies
- Transfers back the remaining balance.
- Chooses the tax rate such that it balances its budget.

$$\underbrace{\left(\int \tilde{p}_o \circ d\omega\right) - \left(\int p_o(b) \circ d\omega\right)}_{\text{Government Spending on Ozempic}} = \underbrace{wh - \tau(wh)}_{\text{Government revenue}} \quad (14)$$

The subsidy schedule is as follows:

$$p_o(b) = \begin{cases} \infty, & \text{if } b < \mathfrak{b} \\ \tilde{p}_o - \iota, & \text{otherwise} \end{cases} \quad (15)$$

# First Order Conditions: FOC( $h$ )

$$V(a, s, d, b, m, e_{-1}) = \max_{\{c, h, x, e, o\}} u(c, x - \underline{x}(d)) - \kappa(b)v(h) - \tilde{n}(e, e_{-1}) \\ + \beta\gamma(a', b', d', m')V(a', s', d', b', m', e)$$

$$\underbrace{u_c(\cdot)\lambda_h}_{\text{Current MB of } h} \underbrace{WS}_{\text{(higher consumption } c)} - \underbrace{\kappa(b)v_h(\cdot)}_{\text{Current MC of } h} \underbrace{(\cdot)}_{\text{(disutility of working)}} + \underbrace{\beta\gamma V_{s'}g_h(\cdot)}_{\text{Future MB of } h} \underbrace{(\cdot)}_{\text{(higher future human capital, } s')}$$

$$+ \underbrace{\left[ -\tilde{n}_e(\cdot) + \beta\gamma[V_{d'}(\cdot)(-\delta_e^{(d)}(\cdot))d + V_{b'}(\cdot)(-\delta_e^{(b)}(\cdot))b + V_e(\cdot)] \right]}_{\text{Future and Current MC of } h} \underbrace{\frac{\partial e}{\partial h}}_{\text{(loss of } e)} = -1$$

$$+ \underbrace{\beta[V(a', s', d', b', m', e)] \left( -\gamma^b(-\delta_e^b b) - \gamma^d(-\delta_e^d d) + \gamma^m(q_e^m) \right)}_{\text{Future MC of } h} \underbrace{\frac{\partial \gamma}{\partial e} \frac{\partial e}{\partial h}}_{\text{(lower utility from survival)}} = 0$$

## First Order Conditions: FOC( $e$ )

$$V(a, s, d, b, m, e_{-1}) = \max_{\{c, h, x, e, o\}} u(c, x - \underline{x}(d)) - \kappa(b)v(h) - \tilde{n}(e, e_{-1}) \\ + \beta\gamma(a', b', d', m')V(a', s', d', b', m', e)$$

FOC( $e$ ):

$$- \underbrace{\tilde{n}_e(\cdot)}_{\substack{\text{Current MC of } e \\ \text{(Disutility of physical exercise)}}} + \beta\gamma \underbrace{\left[ V_{d'}(\cdot)(-\delta_e^{(d)}(\cdot))d + V_{b'}(\cdot)(-\delta_e^{(b)}(\cdot))b + V_e(\cdot) \right]}_{\substack{\text{Future MB of } e \\ \text{(Less } d', \text{ less disutility of working } \kappa(b), \text{ less disutility from } e)}} \\ + \beta \underbrace{\left[ V(a', s', d', b', m', e) \right] \left( -\gamma^b(-\delta_e^b b) - \gamma^d(-\delta_e^d d) + \gamma^m(q_e^m) \right) \frac{\partial \gamma}{\partial e}}_{\substack{\text{Future MB of } e \\ \text{(Higher utility from survival)}}} = 0$$

# First Order Conditions: FOC( $x$ )

$$V(a, s, d, b, m, e_{-1}) = \max_{\{c, h, x, e, o\}} u(c, x - \underline{x}(d)) - \kappa(b)v(h) - \tilde{n}(e, e_{-1}) \\ + \beta\gamma(a', b', d', m')V(a', s', d', b', m', e)$$

FOC( $x$ ):

$$\underbrace{u_c(\cdot)(-p_x)}_{\text{Current MC of } x} + \underbrace{u_x(\cdot)}_{\text{Current MB of } x} + \underbrace{\beta\gamma[V_{d'}(\cdot)q_x^{(d)} + V_{b'}(\cdot)q_x^{(b)}]}_{\text{Future MC of } x \text{ (higher } d' \text{ and higher } b')} \\ + \underbrace{\beta[V(a', s', d', b', m', e)] \left( -\gamma^b(q_x^b) - \gamma^d(q_x^d) \right) \frac{\partial \gamma}{\partial x}}_{\text{Future MC of } x \text{ (lower utility from survival)}} = 0$$

## First Order Conditions: FOC( $o$ )

$$V(a, s, d, b, m, e_{-1}) = \max_{\{c, h, x, e, o\}} u(c, x - \underline{x}(d)) - \kappa(b)v(h) - \tilde{n}(e, e_{-1}) \\ + \beta\gamma(a', b', d', m')V(a', s', d', b', m', e)$$

FOC( $o$ ):

$$\underbrace{u_c(\cdot)(-p_o)}_{\text{Current MC of } o} + \underbrace{\beta\gamma V_{d'}(\cdot)(-\delta_o^{(d)}(\cdot))d}_{\text{Future MB of } o \text{ (lower } d')} \\ + \underbrace{\beta[V(a', s', d', b', m', e)] \left( -\gamma^d(-\delta_o^{(d)})d + \gamma^m(q_o^m) \right)}_{\text{Future MC and MB of } o \text{ (lower utility from survival)}} \frac{\partial \gamma}{\partial o} = 0 \quad (21)$$



## Definition 1. Stationary RCE of the OLG economy

Given initial conditions  $s_0, d_0, b_0, m_0$ , prices  $\{w, p_x\}$ , government policy  $\{\lambda, \iota, b\}$  a RCE is set of, household allocations  $\{c, x, o, e, h, l\}$ , firm Allocations  $\{Y^i, \mathcal{H}^i\}$ , population measures  $\{\Omega\}$ , such that:

1.  $\{c, x, o, e, h, l\}$  solve the household problem,  $\{Y^i, \mathcal{H}^i\}$  solve the firms problem, the governments budget clears.
2. All markets clear.  $\int x d\Omega = Y^x$ ,  $\int o d\Omega = \mathcal{O}$  and the aggregate resource constraint holds  $\int c d\Omega + \int x d\Omega + \int o d\Omega = Y^x + Y^c + \mathcal{O}$
3. The aggregate induced law of motion  $H$  is  $\Omega = H(\Omega)$  where  $\Omega$  is the joint distribution of the population across states  $(a, s, m, b, d, e_{-1})$

A constraint efficient benevolent planner chooses  $\iota$  and  $b$ , maximizing welfare in the economy:

$$\max_{\iota, b} \int V(a, s, d, b, m, e_{-1}) d\Omega \quad (22)$$

s.t

The OLG economy being equilibrium, as per Definition 1. (Stationary RCE)

Choose functional forms and parameters of

$$q^d(x), q^b(x, e, ..), q^m(x, e, ..) \dots \quad (23)$$

$$\delta^d(x), \delta^b(x, e, ..), \delta^m(x, e, ..) \dots \quad (24)$$

$$\gamma(j, b, d, m, \dots) \quad (25)$$

Price schedule of ozempic, wgovy: From prescription regulation and health insurance:

$$p_{o_1}(b)(\omega_{o_1,1}, \omega_{o_1,2}) \quad (26)$$

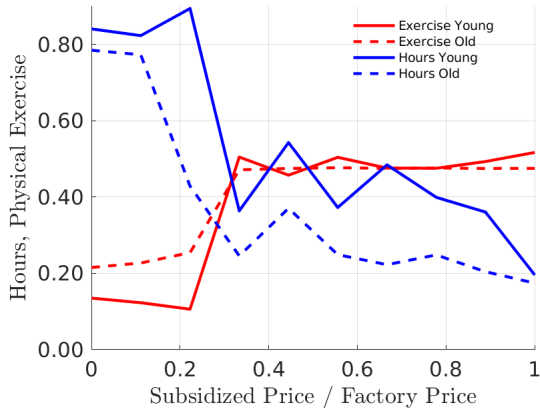
$$p_{o_2}(b)(\omega_{o_2,1}, \omega_{o_2,2}) \quad (27)$$

## Preliminary simulations: Reducing the subsidy

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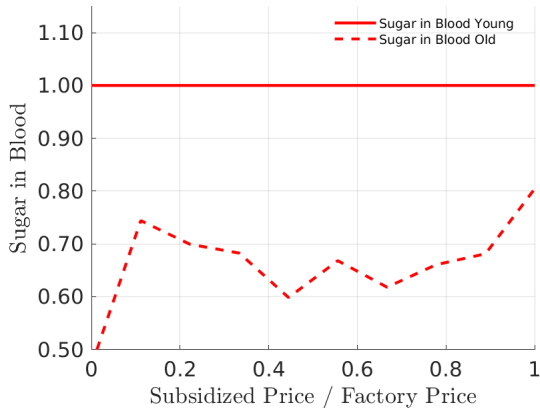
# Reducing Government Subsidies: Hours

Less subsidy reduces working Hours



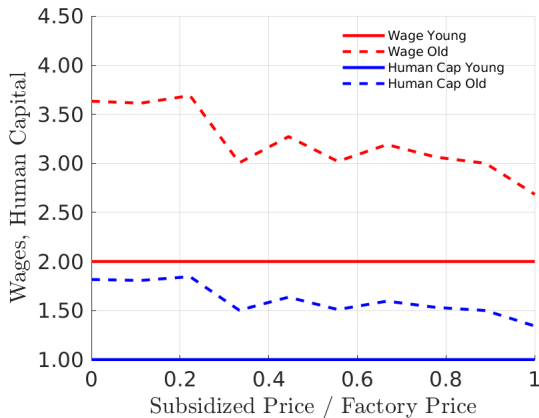
# Reducing Government Subsidies: Sugar in Blood

Less subsidies increase sugar in blood



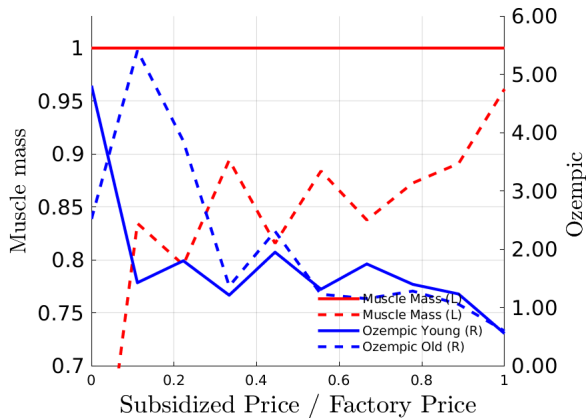
# Reducing Government Subsidies: Wages

Less subsidy reduces wages, though human capital accumulation



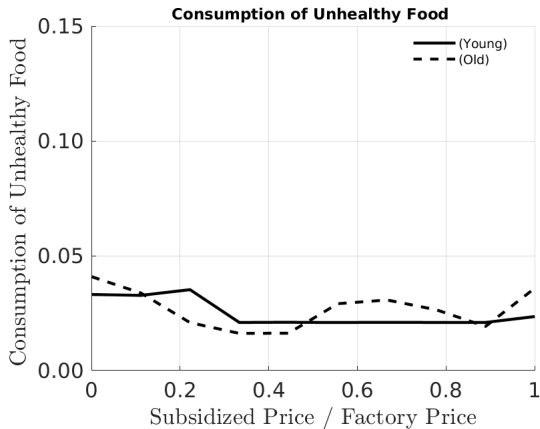
# Reducing Government Subsidies: Muscle mass

Less subsidy reduces drug takeup



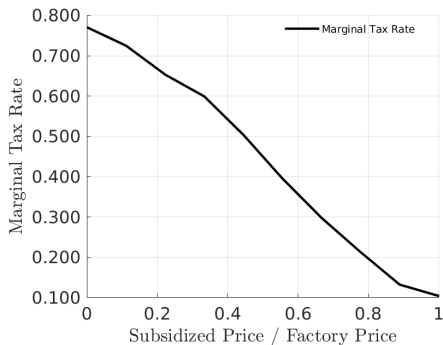


# Reducing Government Subsidies: Consumption of unhealthy food

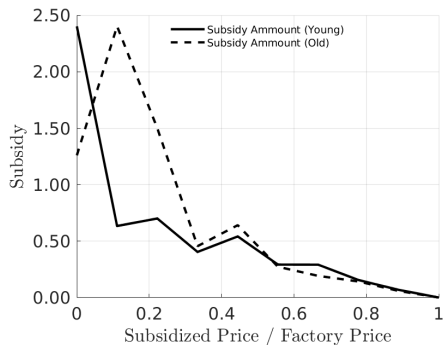


# Reducing Government Subsidies: Tax rate and G. Spnding

(a) Marginal Tax



(b) Government Spending on Ozempic



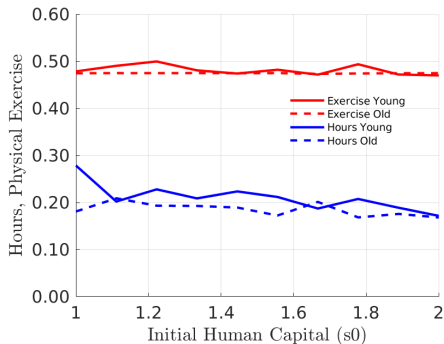
## Ozempic vs Non-Ozempic Economy

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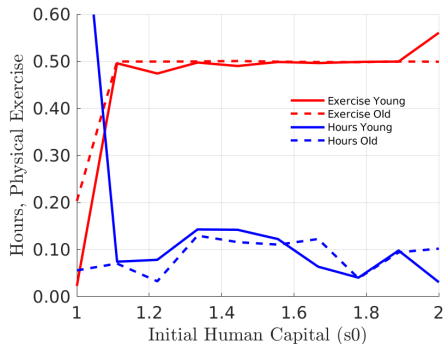
# Work Hours and Physical exercise

## Ozempic increases Hours at the cost of leisure

(a) Subsidized Ozempic

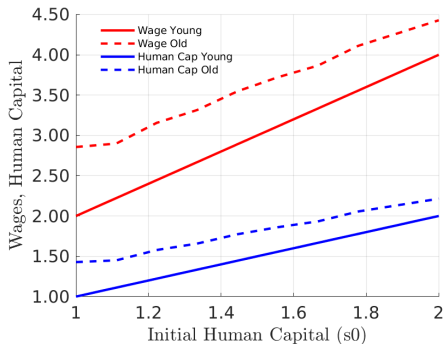


(b) No Ozempic

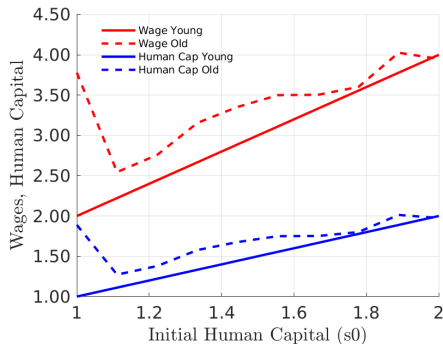


## Ozempic speeds up human capital accumulation thus wage growth

(a) Subsidized Ozempic



(b) No Ozempic



- **Focusing on the diabetic population (Ozempic):**
  - We find that semaglutides:
    - Increase earnings by 2.9%
    - Increase employment by 2.5%
    - Increase working hours by 5%
- **Work in progress:**
  - Empirically
    - ▷ Include the "healthier" non-diabetic population (Wegovy).
  - Quantitatively
    - ▷ Calibration and Optimal Policy

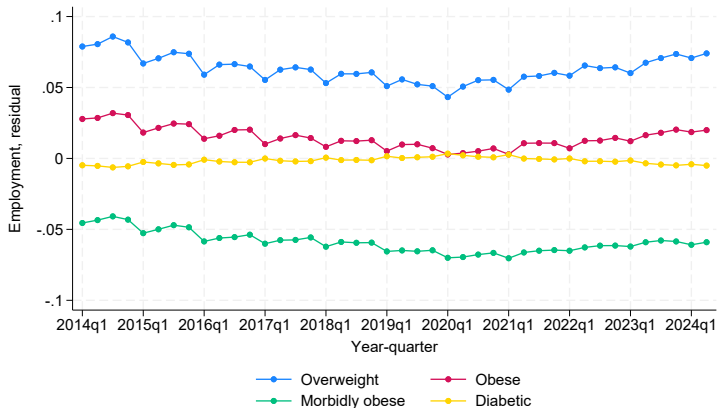
Thank you

# Obesity and Employment (Diabetic sample, Denmark)

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Obesity negative correlated with employment

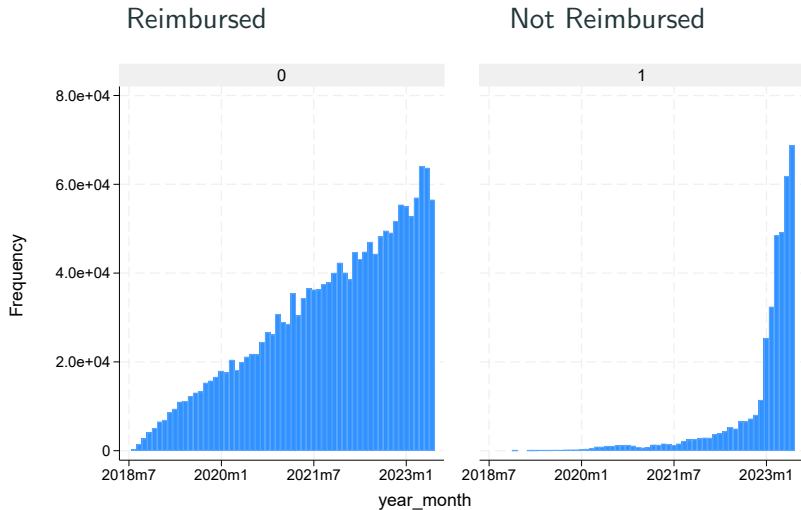
- Among diabetics, morbidly obese are less likely to be employed





# New semaglutide prescriptions (Ozempic, Mounjaro, Wegovy)

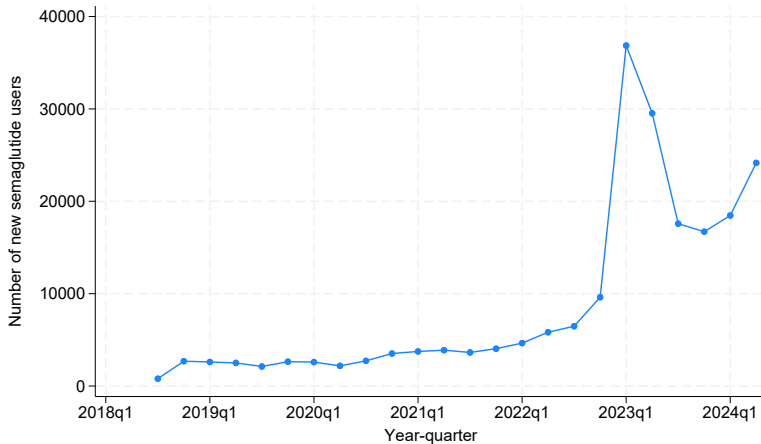
Semaglutide treatment is introduced in 2018.



Graphs by same price

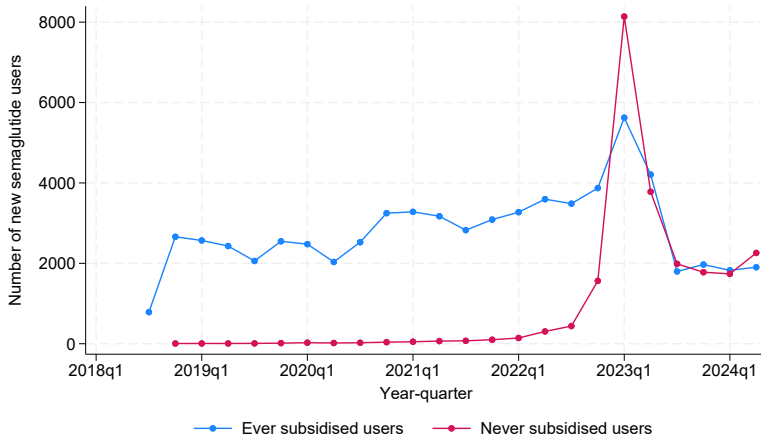
# New semaglutide users

Semaglutide treatment is introduced in 2018.

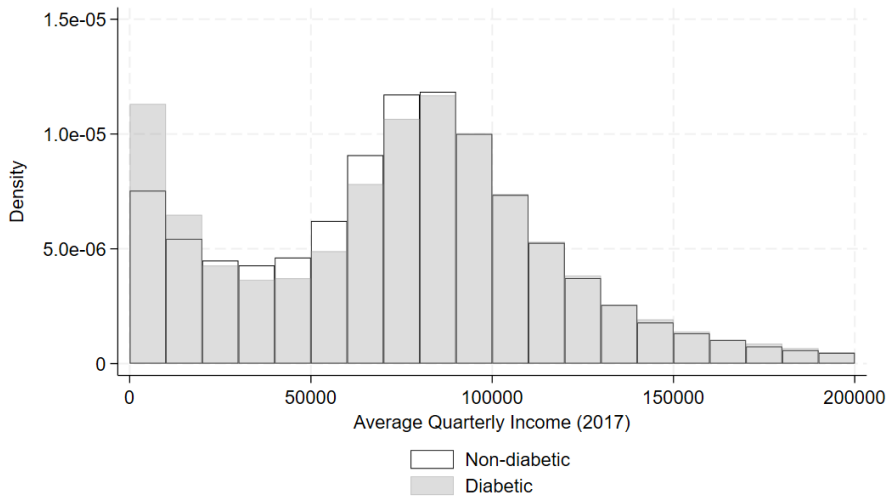


# New semaglutide users: Reimbursement split

Semaglutide treatment is introduced in 2018.



# Income distribution: Overall population vs Diabetic Population



We use two complementary empirical strategies to try and capture the effects of the use of semaglutides on labor market outcomes.

## 1. First prescription of semaglutide vs. First prescription of insulin

- **Treatment** → those prescribed a semaglutide for the first time, with no previous insulin prescriptions.
- **Control** → those prescribed insulin for the first time, with no prior prescriptions of insulin or semaglutides.

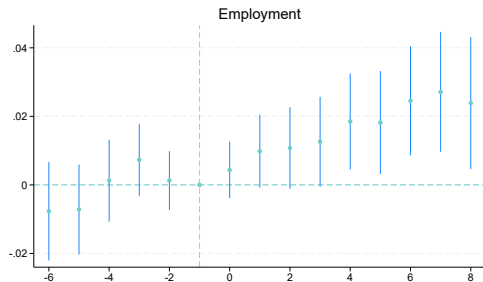
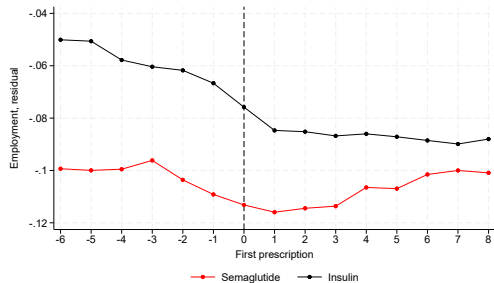
# Insulin vs. Semaglutides - Empirical Strategy

We use a difference-in-difference event study set-up:

$$y_{it} = \sum_{\tau=-6, \tau \neq -1}^8 \gamma_{\tau} D_{i\tau(it)} + \mu_{\tau(it)} + \nu_t + X_{it} + \epsilon_{it}$$

- $Y_{it}$ : employment, log earnings for individual  $i$  in quarter  $t$
- $D_{i\tau(it)}$ : treatment (first semaglutide prescription), control (first insulin prescription) in quarter  $\tau$
- $\mu_{\tau(it)}$  is the time relative to first treatment fixed effect
- $\nu_t$ : calendar year and quarter fixed effects.
- $X_{it}$ : individual controls

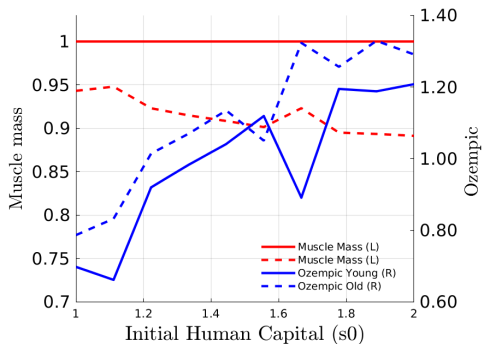
# Event Study: Insulin vs. Semaglutides



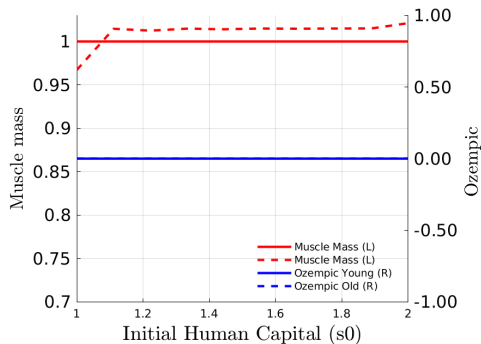
# Muscle Mass:

## Ozempic reduces muscle mass, offsetting exercise

(a) Subsidized Ozempic



(b) No Ozempic

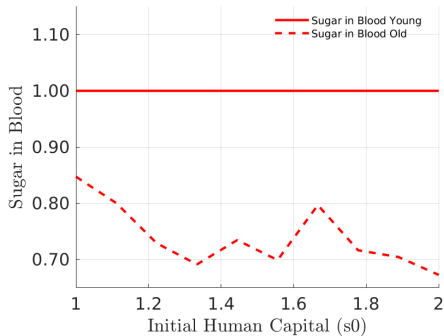




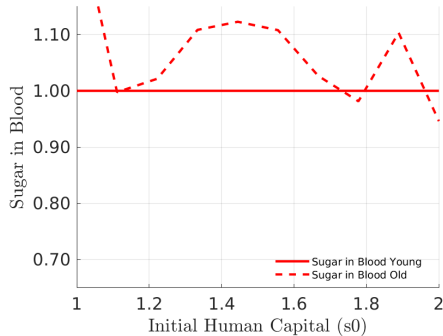
# Sugar in Blood:

Ozempic reduces reduces sugar in blood

(a) Subsidized Ozempic



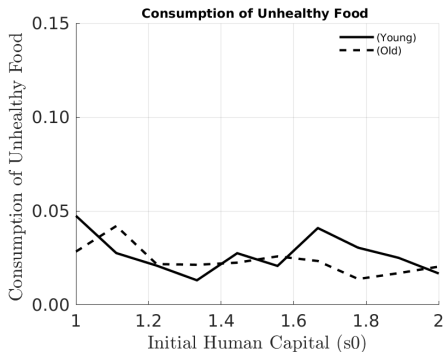
(b) No Ozempic



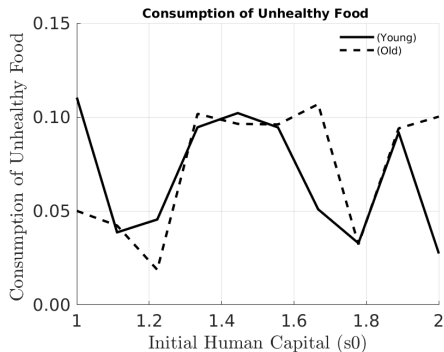
# Consumption of unhealthy food:

## Ozempic reduces reduces the consumption of unhealthy food

(a) Subsidized Ozempic

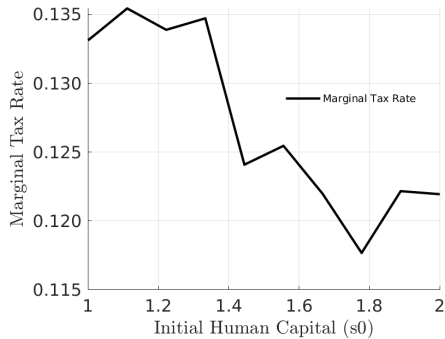


(b) No Ozempic

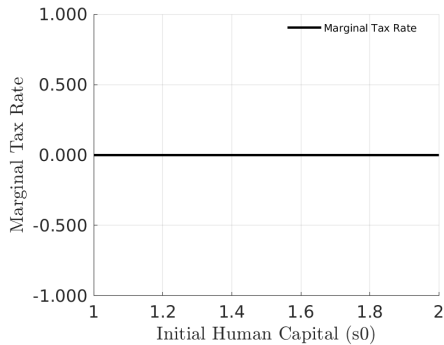


# Tax Rate

(a) Subsidized Ozempic



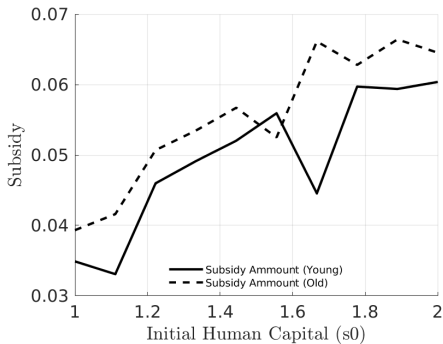
(b) No Ozempic



# Government Spending

## Ozempic spending increases with wealth

(a) Subsidized Ozempic



(b) No Ozempic

